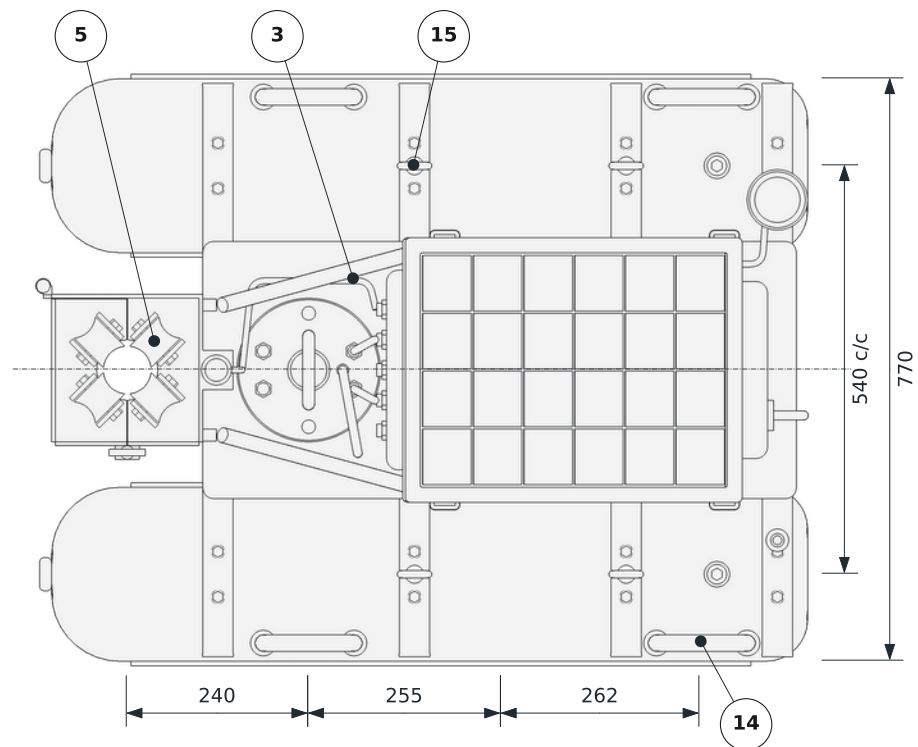
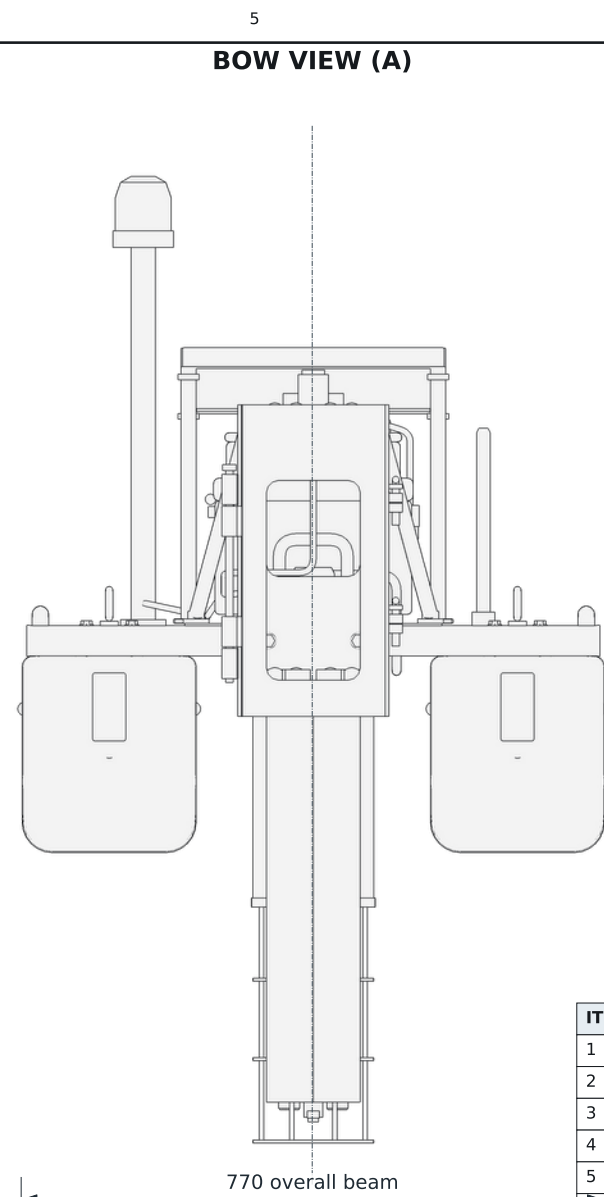
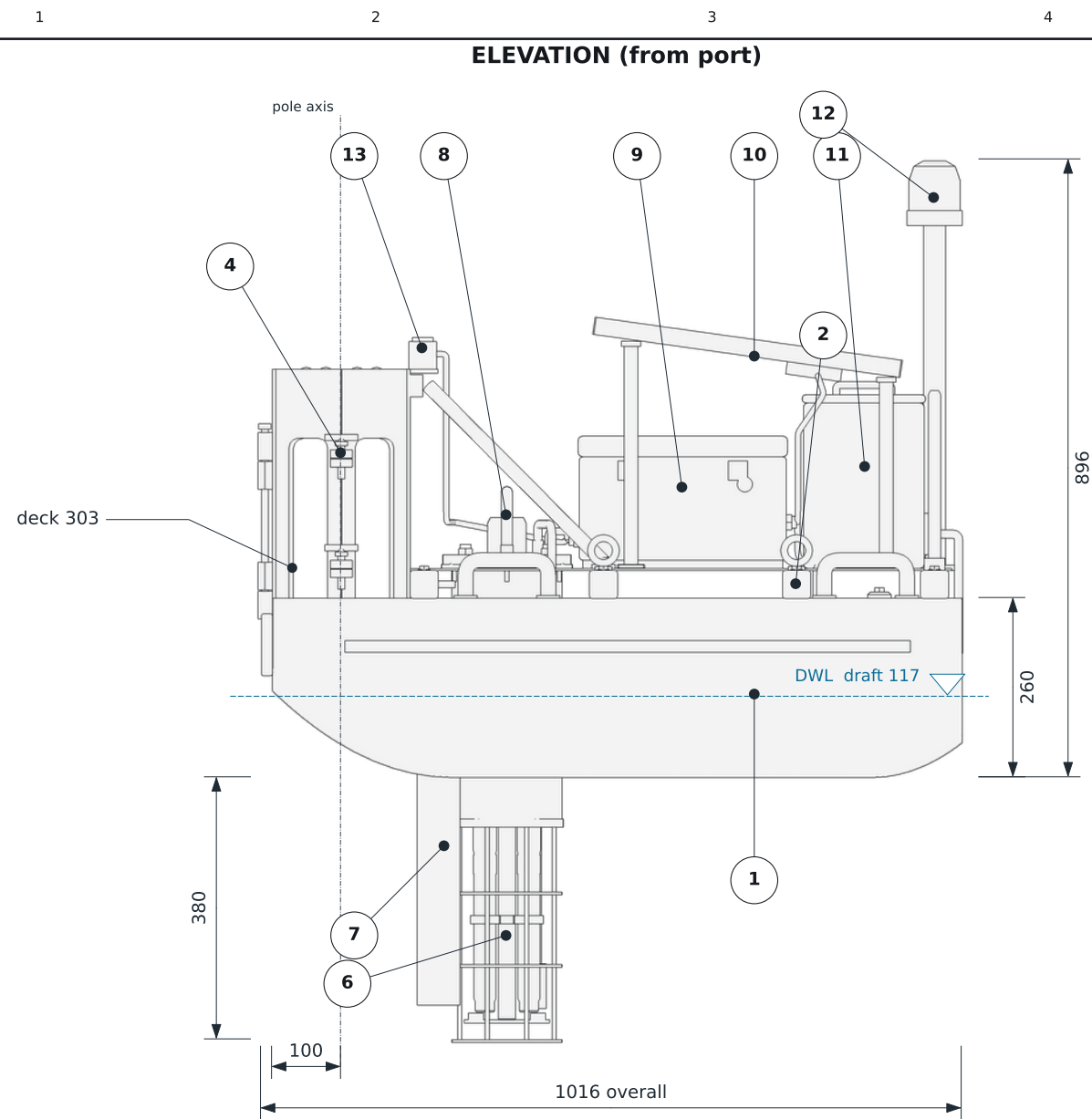




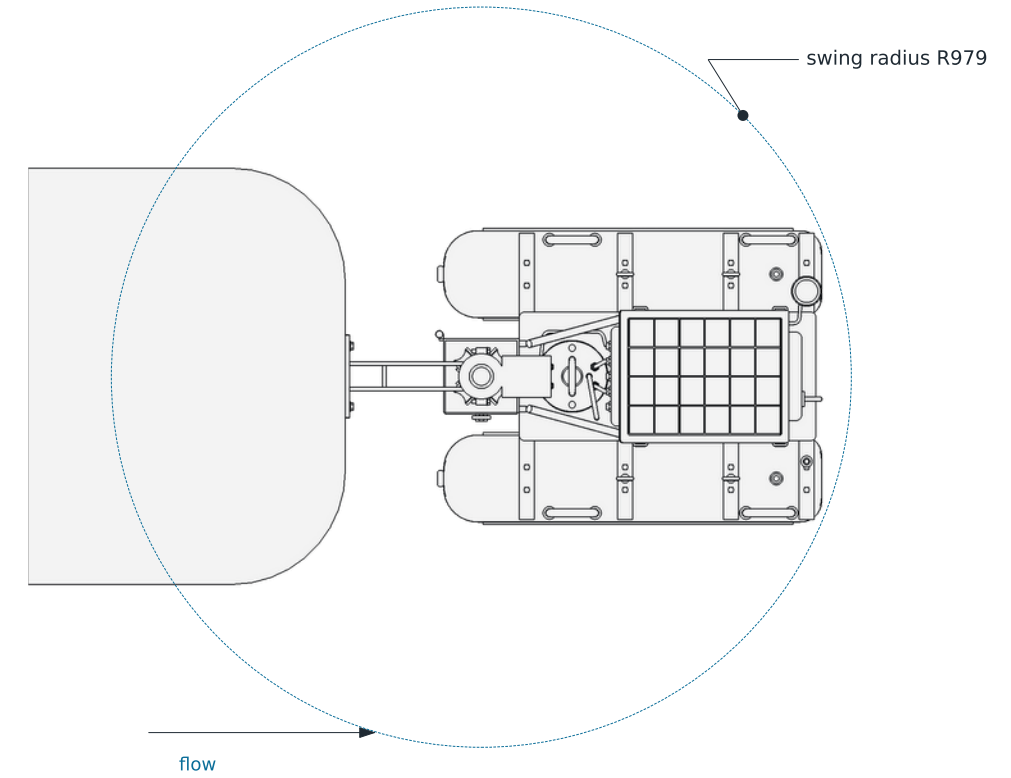
ITEM	PART No.	DESCRIPTION	QTY
1	AS3-201	Pontoon hull, rotomoulded PE, foam filled	2
2	AS3-211	Crossbeam 6061 SHS 40x40x3	4
3	AS3-213	Equipment deck plate 6061 t3	1
4	AS3-301/302	Guide carriage (fixed half + gate)	1
5	AS3-311	Concave guide roller UHMW-PE	8
6	AS3-401/403	Sensor well + guard cage	1
7	AS3-404	V debris deflector HDPE	1
8	AS3-411..417	Pull-up sensor cartridge (4 probes)	1
9	AS3-511	Power and control pod IP67	1
10	AS3-521	PV module 20 W on hinged canopy	1
11	AS3-601	Autosampler cassette (4 x 250 mL)	1
12	AS3-703	Jal-Deep 360 deg beacon	1
13	AS3-712	Stage sensor (ultrasonic, up)	1
14	AS3-801	Grab handle	4
15	AS3-802	Lifting eye M12	4

NOTES

1. Mass from CAD 48.54 kg (sampler bottles full); draft 116.8 mm, freeboard 143.2 mm.
2. Reserve buoyancy 2.27x; GM_T 0.551 m; GZ positive to 60 deg (free floating).
3. Pole sits 100 mm inside the bow line between the hulls; float trails downstream (weathervane).
4. All fasteners A4-70 stainless; nylon isolation at every 316/6061 joint.
5. Full BOM: cad/BOM_R3.csv. Every part has a STEP file in cad/parts/.

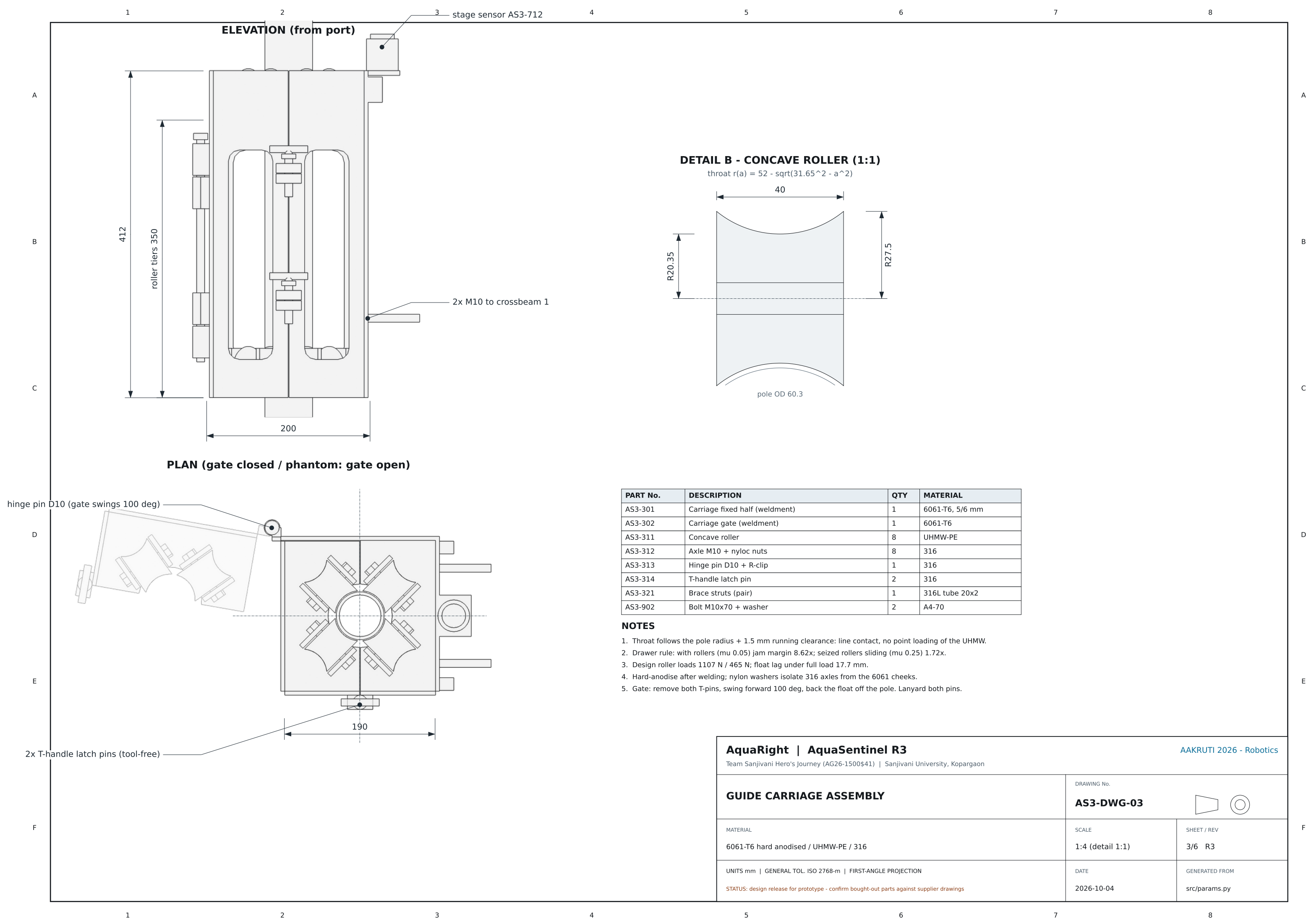
AquaRight AquaSentinel R3 Team Sanjivani Hero's Journey (AG26-1500\$41) Sanjivani University, Kopergaon		
GENERAL ARRANGEMENT - FLOAT	DRAWING No.	
	AS3-DWG-01	
MATERIAL	SCALE	SHEET / REV
As parts list	1:10	1/6 R3
UNITS mm GENERAL TOL. ISO 2768-m FIRST-ANGLE PROJECTION	DATE	GENERATED FROM
STATUS: design release for prototype - confirm bought-out parts against supplier drawings	2026-10-04	src/params.py

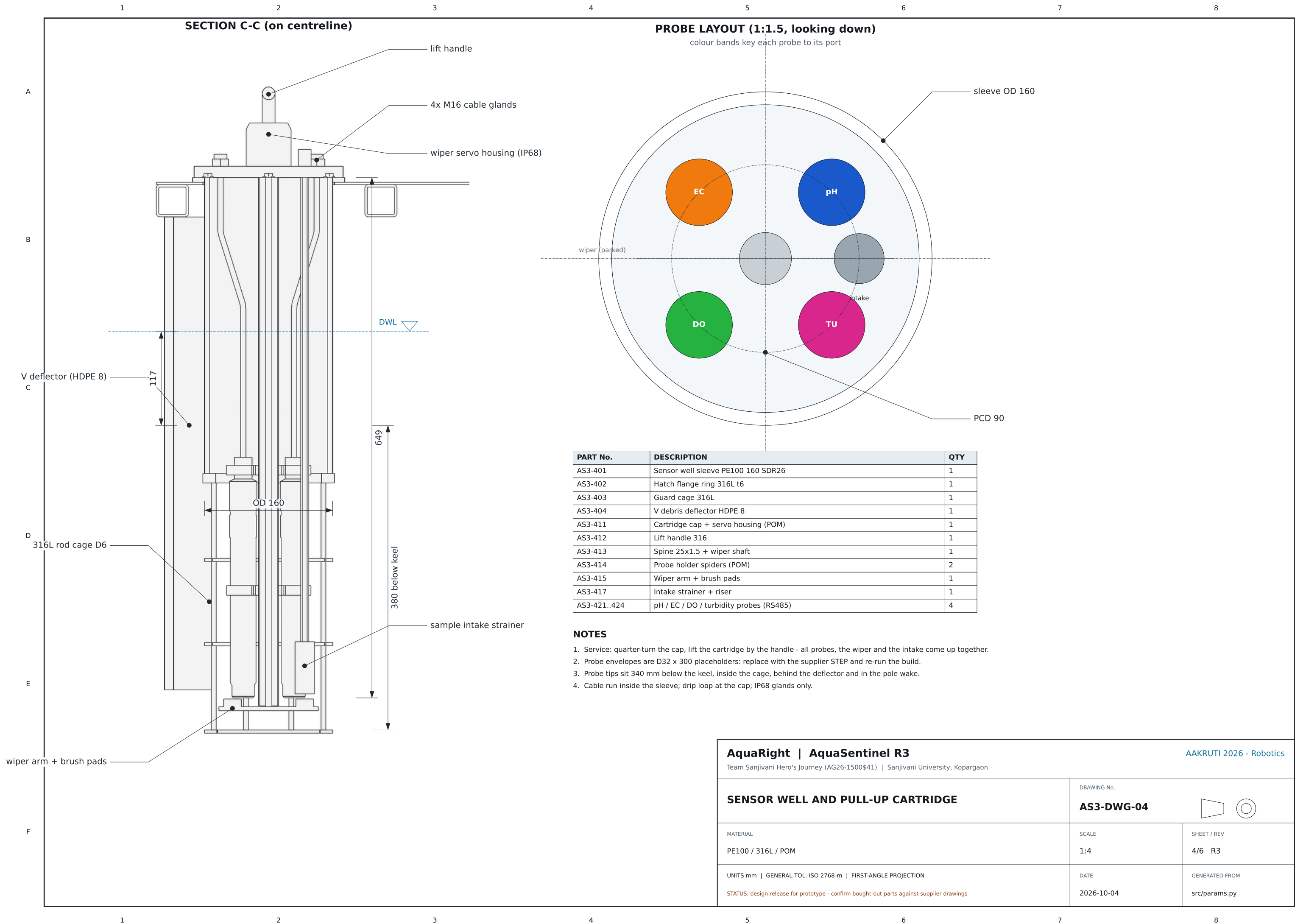
PLAN - free yaw about the pole (weathervane)



1. Brackets are identical (AS3-101). Install the lower one at or below the lowest recorded level (-47 on this datum).
2. Anchors: 4x M12 A4 chemical anchors per bracket into sound pier concrete; confirm pull-out with the pier owner / WRD.
3. Design stage range 2000 mm is an ASSUMPTION: set from the site gauge record before cutting the pole.
4. Float rides on 8 rollers between the stops; at the stops it is held, never jammed (EPDM buffers).
5. Free yaw: the float trails downstream. Only the hull rub rails and bow bumpers can touch the pier.
6. Beyond 2 m stage or 1.5 m/s current: park at the top stop or open the gate and retrieve.
7. Pole loads (1.5 m/s + 500 N debris): 54.8 MPa pinned, SF 3.1 vs 170 MPa yield.

AquaRight AquaSentinel R3			AAKRUTI 2026 - Robotics		
Team Sanjivani Hero's Journey (AG26-1500\$41) Sanjivani University, Kopergaon					
INSTALLATION AND STAGE ENVELOPE			DRAWING No.		
			AS3-DWG-02		
MATERIAL			SCALE		SHEET / REV
316L / concrete pier			1:25		2/6 R3
UNITS mm GENERAL TOL. ISO 2768-m FIRST-ANGLE PROJECTION			DATE		GENERATED FROM
STATUS: design release for prototype - confirm bought-out parts against supplier drawings			2026-10-04		src/params.py





A
B
C
D
E
F

1

2

3

4

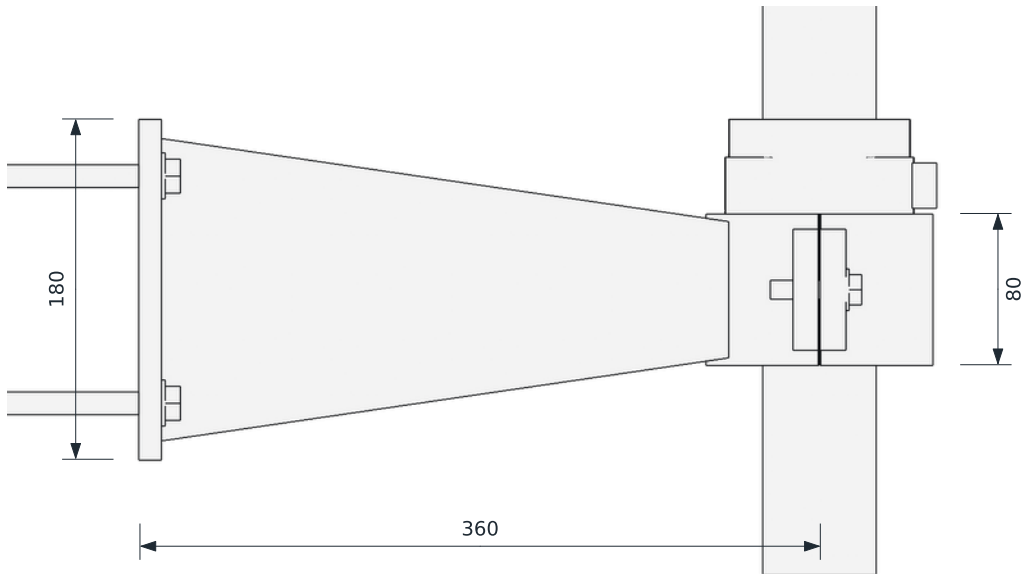
5

6

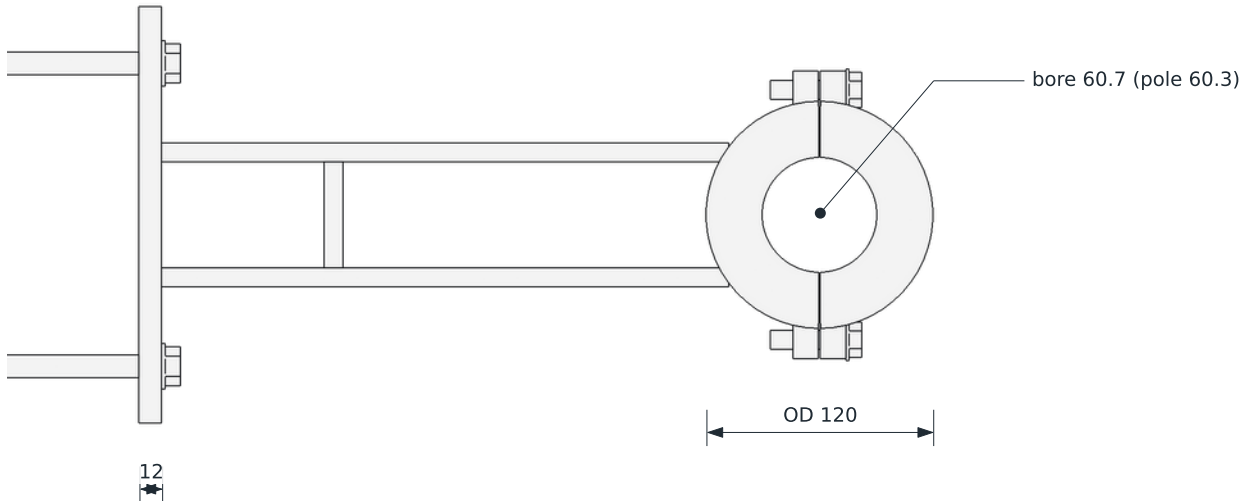
7

8

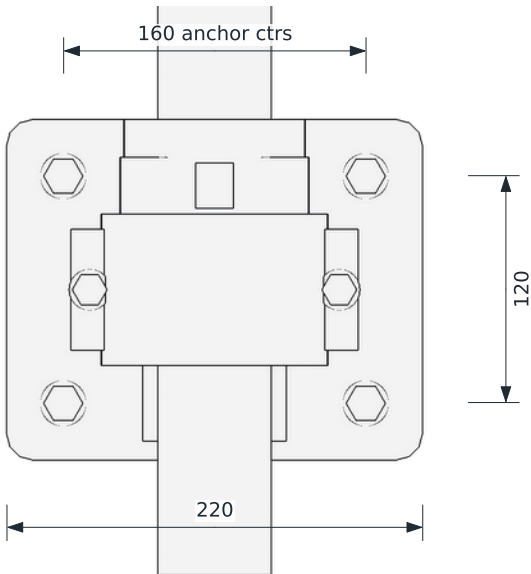
ELEVATION



PLAN



VIEW FROM RIVER (B)



PART No.	DESCRIPTION	QTY	MATERIAL
AS3-101	Pier bracket weldment	2	316L 10/12 mm
AS3-102	Clamp cap	2	316L
AS3-103	Stage target plate (upper only)	1	316L t4
AS3-905	Bolt M10x40 + washer	4	A4-70
AS3-904	Chemical anchor M12x130	8	A4 + mortar
AS3-113/114	Stop collar + EPDM buffer	2+2	316L / EPDM

NOTES

- Upper and lower brackets are identical: one weld fixture, one drawing.
- Design reaction per bracket 430.0 N horizontal (1.5 m/s current + 500 N debris).
- Anchor embedment and edge distance to the anchor maker's data for the pier concrete grade - verify on site.
- Weld: TIG, ER316LSi filler, full fillet all round; pickle and passivate after welding.
- Clamp bore reamed 60.7; tighten M10 to 35 N.m; pole must not rotate or slip under hand load.

AquaRight | AquaSentinel R3

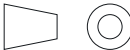
Team Sanjivani Hero's Journey (AG26-1500\$41) | Sanjivani University, Kopergaon

AAKRUTI 2026 - Robotics

PIER BRACKET (WELDMENT) + CLAMP CAP

DRAWING No.

AS3-DWG-06



MATERIAL

SS 316L, passivated

SCALE

1:4

SHEET / REV

6/6 R3

UNITS mm | GENERAL TOL. ISO 2768-m | FIRST-ANGLE PROJECTION

STATUS: design release for prototype - confirm bought-out parts against supplier drawings

DATE

2026-10-04

GENERATED FROM

src/params.py

1

2

3

4

5

6

7

8

A
B
C
D
E
F